

AI Engineering (Week 1)

aie26

Topic 02 : Motivating Example

Part 01 : Introduction to Classification

Preparation

Dr Bernard Butler

Department of Computing and Mathematics, Waterford Institute of Technology.

(bernard.butler@setu.ie)

Data Handling

Exploring Data

Exploring Data 2

Building Models

Autumn Semester, 2026

Outline

- How classification differs from regression
- Classification metrics
- Lazy vs Eager learners

Wrap up

AI Engineering (Week 2)

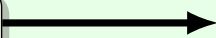
Introduction



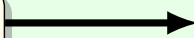
Motivating Example

Preparation

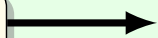
Data Handling



Exploring Data 1



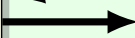
Exploring Data 2



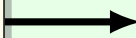
Building Models

Prediction

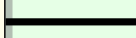
Regression
1



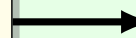
Classification
1



Regression
2

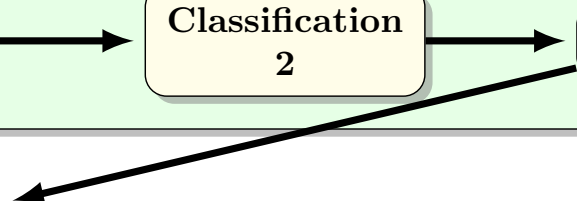
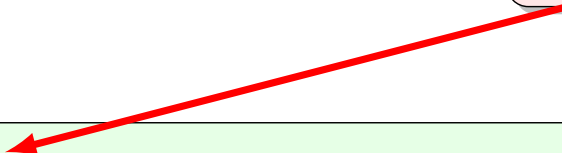


Classification
2



Clustering

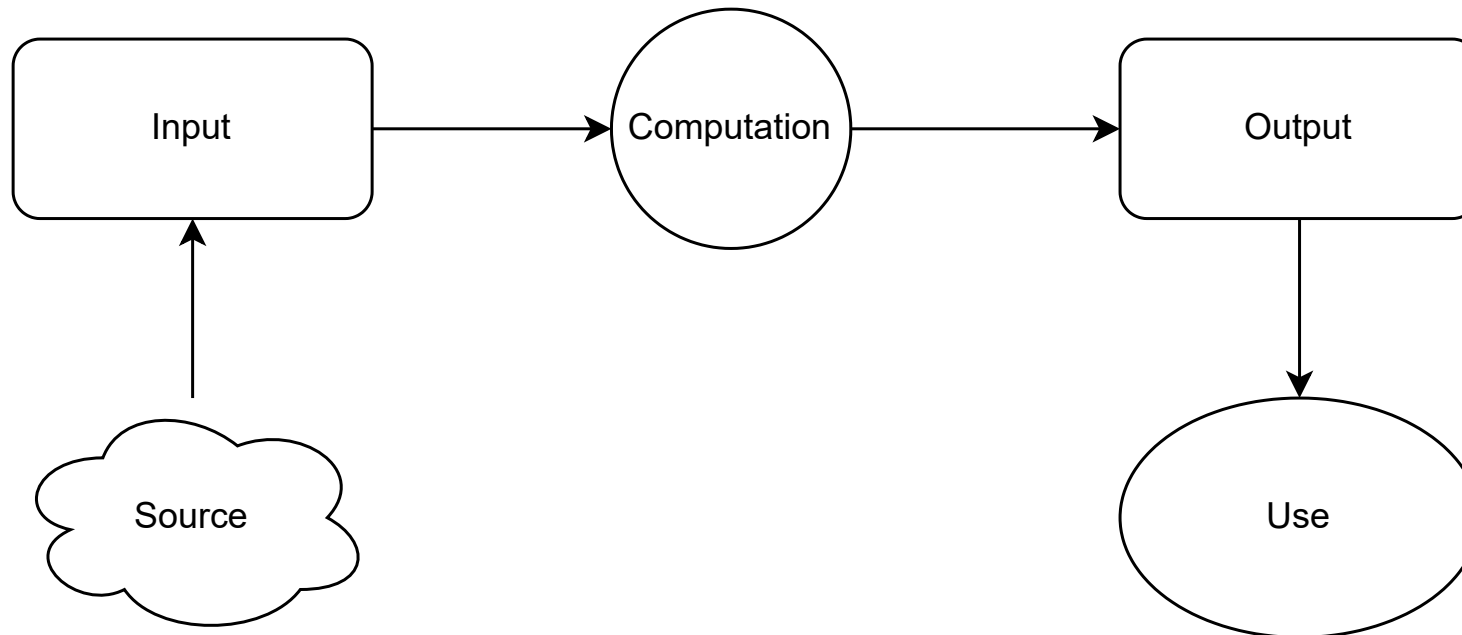
Wrap up



Outline

1. Introduction	3
1.1. Learning from data	4
1.2. Lazy vs Eager Learners	6
2. Introduction to Classification	7
3. k Nearest Neighbours	11
4. Summary	22

What does it mean to learn from data?



Programmed Computation

- Explicit, detailed programming logic
- Handle edge cases, messy data
- Software engineering - unit testing, etc.

Learned Computation

- Implicit, learning from examples
- Less brittle, but harder to test
- New paradigms - iterative model building

Machine learning pros and cons

Benefits

- Less programming effort
- Subtle rules are inferred
- One algorithm works for a range of problems
- Most of the code is in libraries
- Scales better with data complexity

Challenges

- Need (lots of) data for training
- Training data (sample) needs to represent population
- Algorithms have many configuration settings
- Need to understand and validate
- Prediction error needs to be minimised

Lazy vs Eager Learners

Lazy learner

Stores training data (or only minor processing) and uses local approximation to predict a value from test data.

- Does not generalise until after training
- Does not produce a standalone model
- Training data must be kept for prediction
- Local approximations
- Often based on *search*
- If new data is just added to the training data, it can respond more easily to changing conditions

Eager learner

Builds a model from the train set, before receiving new data for prediction

- Training has an extra goal: to generalise from the data
- Training has an extra output: standalone model
- Training data can be discarded after use
- Local and/or global approximations
- Based on *computation*
- Models *drift* with time, so not suited to highly dynamic contexts, as it needs retraining

Usually an (eager) model requires much less memory than a (lazy) training set.

Introduction to Classification

Definition 1 (Classification)

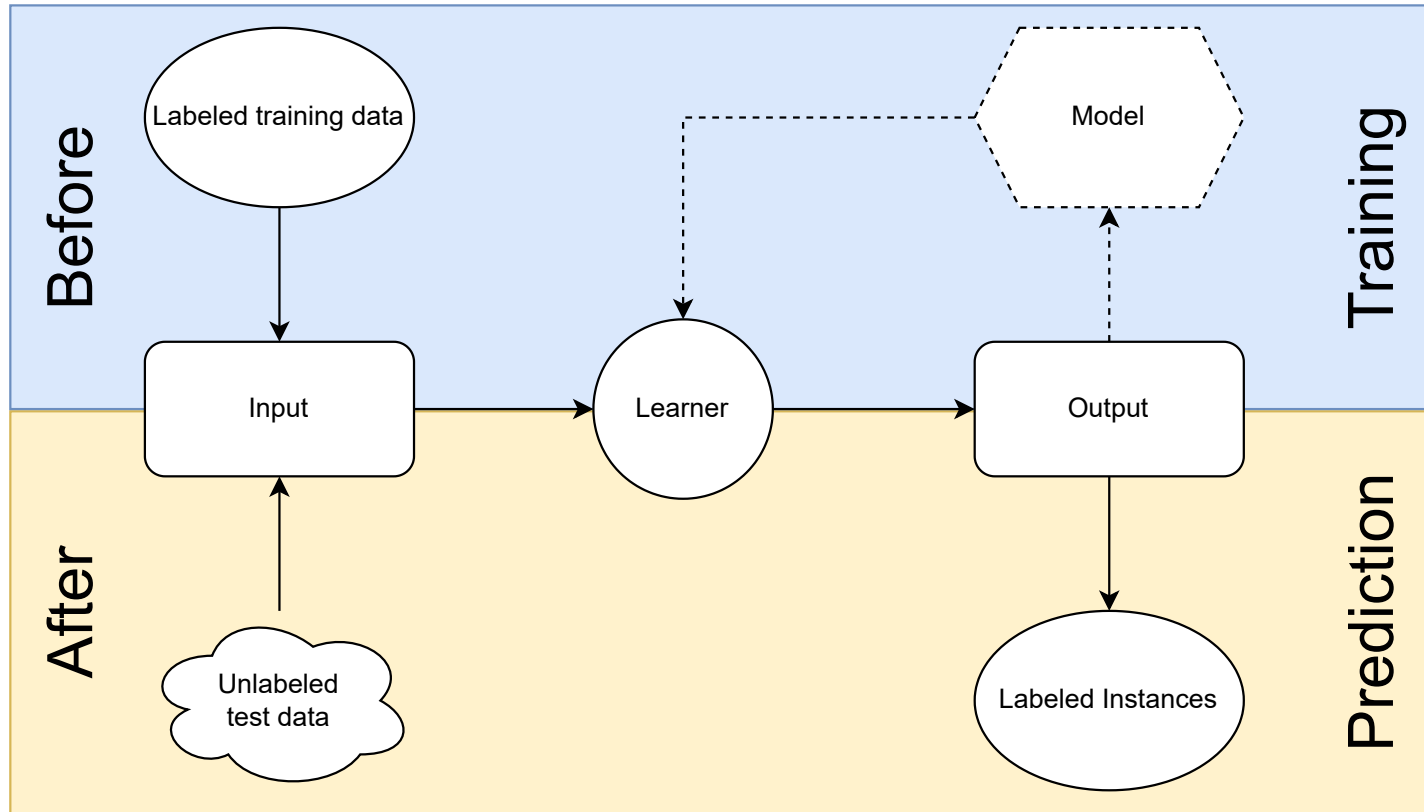
Classification aims to learn a function that takes attribute values and predicts a categorical/qualitative value, such as membership of a class, existence of an effect, etc.

The attributes can be categorical or numeric.

Classification is an example of *supervised learning* because it requires a training set of labeled observations.

- Some *classifiers* generate class membership probabilities en route to predicting class membership (of the most likely class), so the predicted class can be defined by a set of numbers rather than a simple label.
- There are many classification algorithms!
- We choose one of the simplest today, which works by *voting for the most likely label*.

Classification Overview



Today's classifier is a lazy learner and so uses local approximation, not a model

Example Applications

➤ In 5 minutes, identify 3 possible applications for classification ➤

Motivation

Example 2 (Spam Detection)

A new email arrives. Is it spam? We have a large database of previous emails that have been labeled “Spam” or “Ham”. Can we use this information *directly* to say whether the new email is spam or not?

Given each of the following

- ① database of n instances $\{x_i\}$ with p attribute values per instance
- ② distance function $D : d(x_i, x_j) : \mathbb{R}^{p \times p} \rightarrow \mathbb{R}$ where $d(x_i, x_j) > 0$ if $x_i \neq x_j$ and is zero otherwise
- ③ function S that searches for instances that “match” an incoming instance based on D
- ④ function R that identifies the k “nearest” (as defined by D) instances
- ⑤ function A that aggregates the “labels” of these k neighbours, yielding one representative value
- ⑥ function L that applies this representative label to the incoming instance



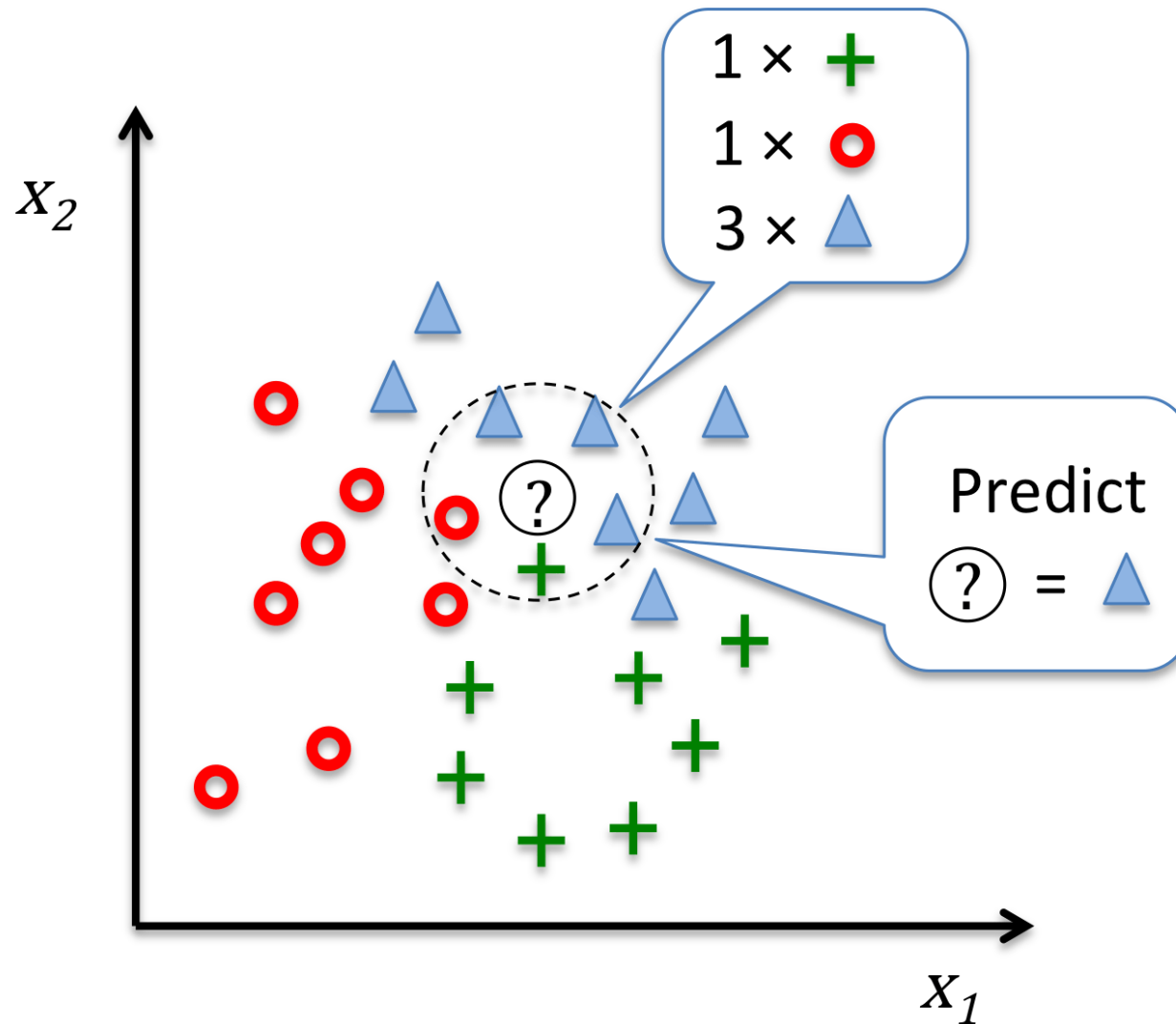
K-Nearest Neighbours: Practical Considerations

Implementation

- 1 The training set needs to be stored in a format (such as a `pandas` dataframe) that is ready for both searching and computation
- 2 The distance function D needs to take account of all the relevant dimensions/attributes, possibly weighted
- 3 The search S and ranking R functions needs to work well together
- 4 The aggregation function A for k -nearest neighbours just takes the most frequent value (also known as the *mode*) of the k existing labels

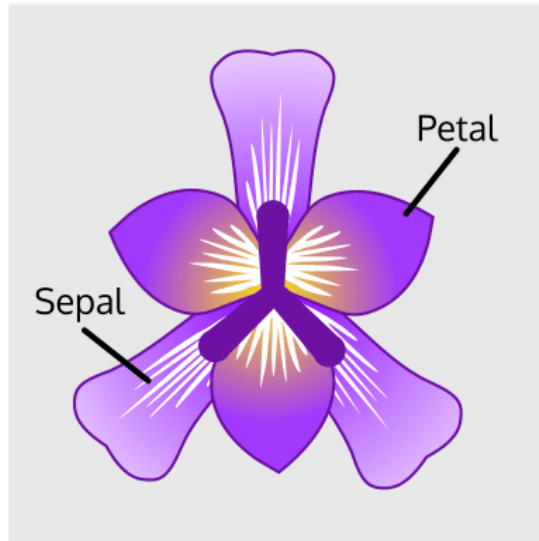
Conceptually this is a very simple algorithm. It can be tweaked by varying k and D (or, very rarely, A). Implementations exist in python (in `scikit-learn`).

K-Nearest Neighbours: Example prediction

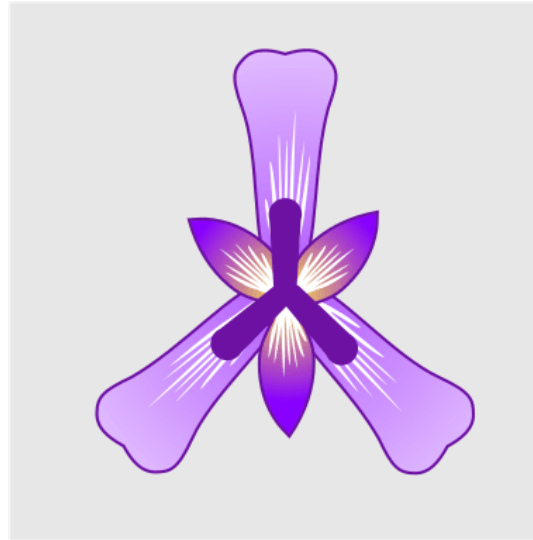


“Follow the crowd”: the new instance is labeled with a blue triangle

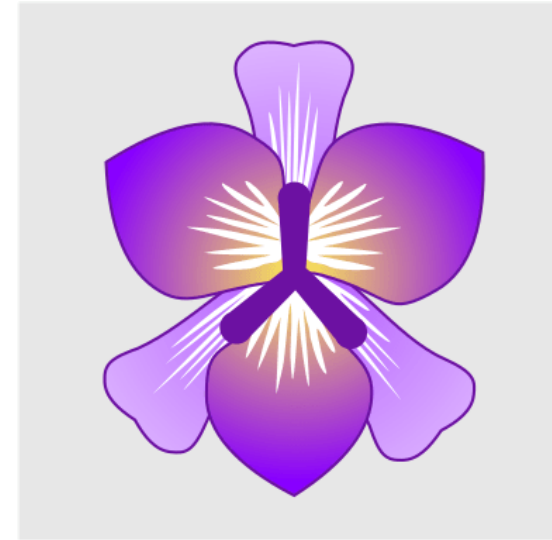
Classifying iris species



Iris Versicolor



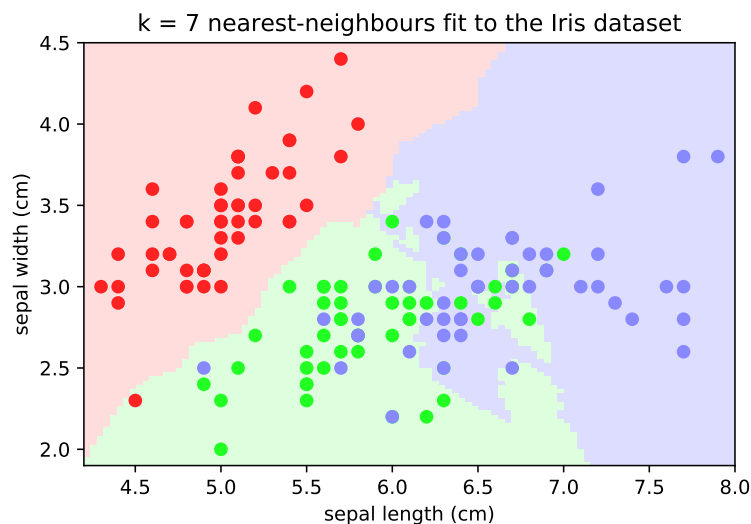
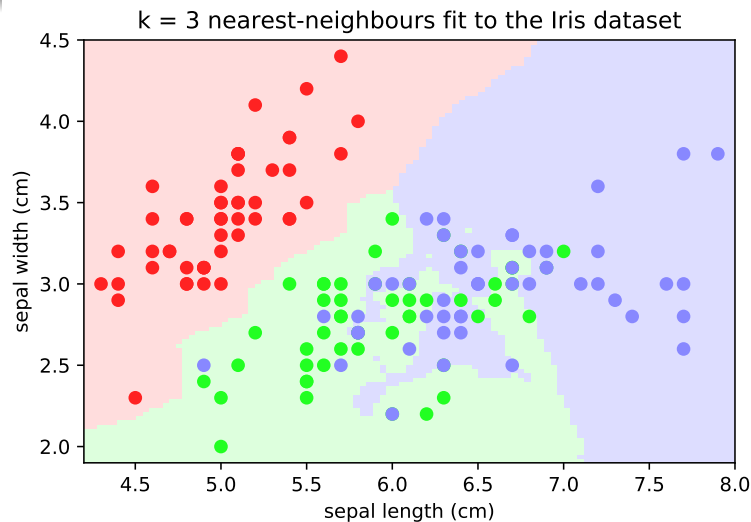
Iris Setosa



Iris Virginica

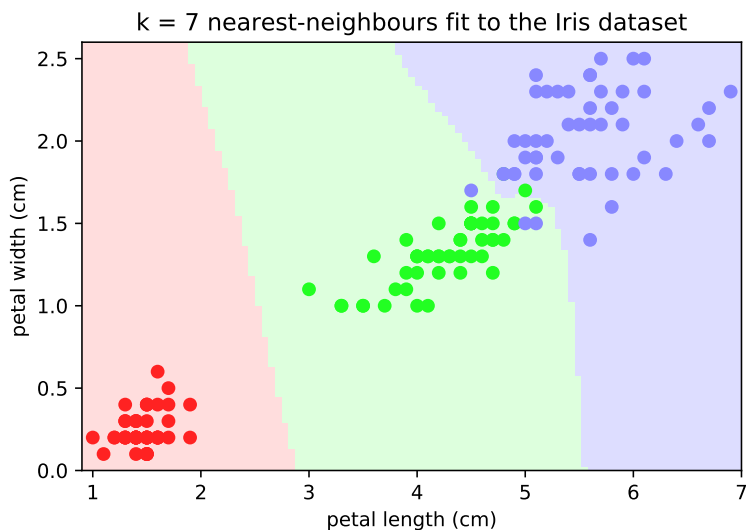
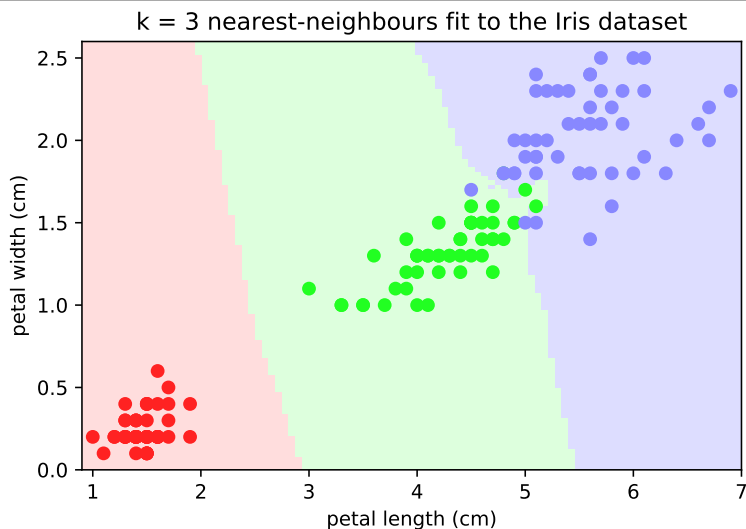
Given measurements of sepal and petal lengths and widths, can we distinguish between the 3 species?

K-Nearest Neighbours: Iris SW-SL



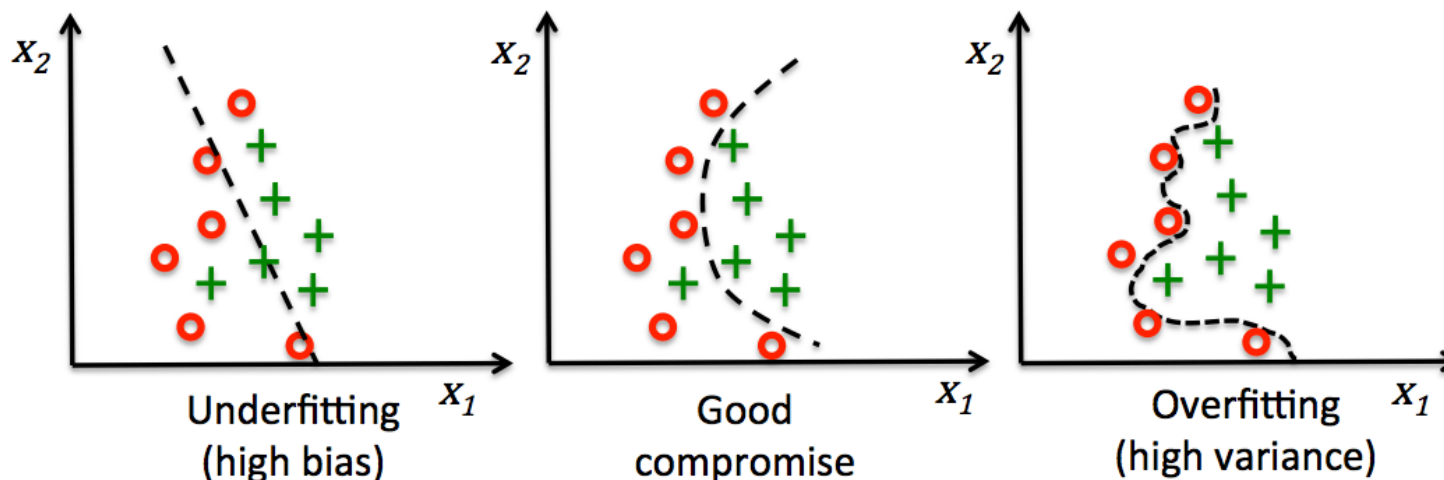
- The Iris dataset has 4 descriptive attributes, so there are 6 possible pairs
- Of these, the Sepal-Width $\times$ Sepal-Length combination is the least effective at distinguishing between the three species
- In this plot, *I. setosa* (red) is well separated from *I. versicolor* (green) and *I. virginica* (blue)
- However the boundary between *I. versicolor* (green) and *I. virginica* (blue) is unclear
- $k = 3$ has relatively low bias and (possibly) high variance
- $k = 7$ has lower variance, pays less attention to “outliers”, so region boundaries are smoother

K-Nearest Neighbours: Iris PW-PL



- As can be seen, the Petal-Width $\times$ Petal-Length combination separates Iris species better
- There are still some difficulties distinguishing between *I. versicolor* (green) and *I. virginica* (blue).
- The size of k does have some effect, but not as dramatically as the more difficult SW-SL combination
- The distance function D depends on the number of dimensions p
- If the regions are well separated, as here, adding more dimensions rarely helps
- Over- and under-fitting is largely down to the choice of k

Sidebar: Classification over- and under-fitting



Generally, under-fitted models do not follow the **training** set closely enough, and so are likely to miss comparable features in the **test** set.

Over-fitted models do the opposite, pay too much attention to peculiarities of the **training** set. They “wiggle” too much!

Setting $k = 1$ ensures that all the training data is correctly labeled (by definition) but it rarely generalises well.

As k increases the boundary becomes smoother. Often that is what you need.

Sidebar: Classification result summary: Confusion Matrix

k = 1, training

predicted		Actual		
		S	V1	V2
predicted	S	50	0	0
	V1	0	50	0
	V2	0	0	50

Each instance is assigned the correct label. There are no off-diagonal terms. **S** represents *I. setosa*, **V1** represents *I. versicolor* and **V2** represents *I. virginica*.

k = 3, training

predicted		Actual		
		S	V1	V2
predicted	S	50	0	0
	V1	0	47	3
	V2	0	3	47

Each training instance of *I. setosa* is assigned the right label. However, of the 50 each of *I. versicolor* and *I. virginica*, 3 of each were incorrectly predicted to be the other.

k = 3, test

predicted		Actual		
		S	V1	V2
predicted	S	10	0	0
	V1	0	7	3
	V2	0	0	10

Each test instance of *I. setosa* and *I. virginica* is assigned the right label. However, of the 10 predicted *I. versicolor* (from a stratified sample), 3 were actually *I. virginica*.

- k-nearest neighbours works better with “small” dimension p but can scale well to “large” number of cases n .
- Unlike most other techniques, decision boundaries are implicit, not explicit.
- If the decision boundary is also needed, might be better to use a different algorithm.

k-nearest-neighbours in python

Python's `scikit-learn` libraries provide a general interface to model fitting that abstracts away most of the details.

Method (Identifying the Iris species)

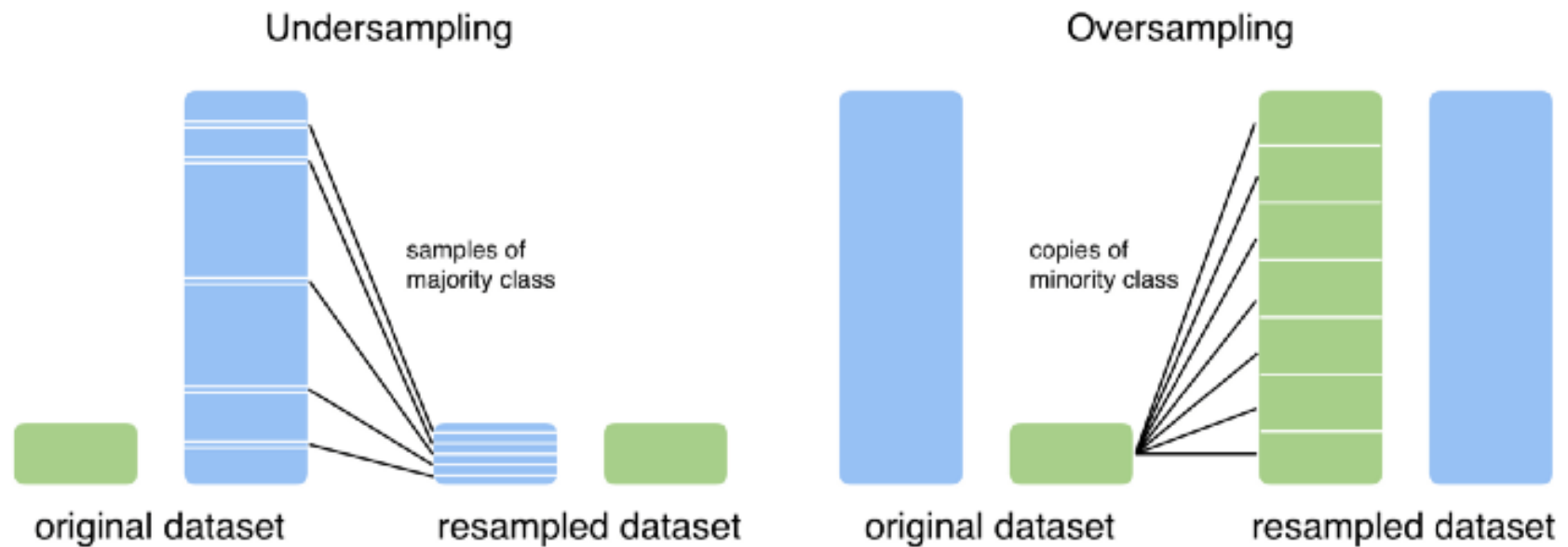
```
1  # create the model
2  knn = neighbors.KNeighborsClassifier(n_neighbors=5)
3
4  # fit the model
5  knn.fit(X, y)
6
7  # What kind of iris has 3cm x 5cm sepal and 4cm x 2cm petal?
8  result = knn.predict([[3, 5, 4, 2],])
9
10 # it is a versicolor...
11 print(iris.target_names[result])
12
13 # class membership probabilities are [0. , 0.8, 0.2]
14 knn.predict_proba([[3, 5, 4, 2],])
```

Very few lines of code are needed!

Have we chosen the best model form?

- There are many classifiers, with many having different “flavours”!
- We have seen just one today (KNN).
- How could we compare different classifiers?

Are the target classes balanced?

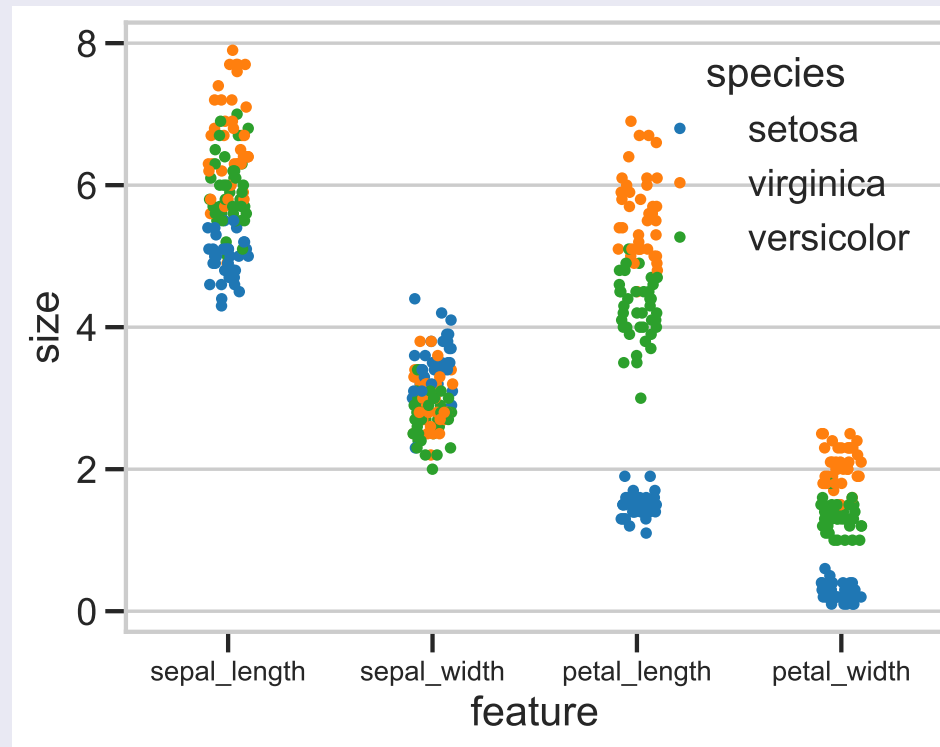


➤ K nearest neighbors is very sensitive to unbalanced data, so need to be careful!!

How might things go wrong?

Have we chosen the best (most predictive) features?

What does this stripplot tell us?



```
ax = sns.stripplot(x = 'feature', y = 'size', hue = 'species', data = data_train_long)
# Reduce the size of the x-tick labels, so they fit better
ax.tick_params(axis="x", labelsize=12)
# Now show the resulting plots.
plt.show()
```

Have we chosen the best (most predictive) k ?

- What values of k can we choose from?
- How can we compare different values of k ?
- How do we search for the best k ?

Summary

- Classification is a *supervised learning* operation, where training data is used to configure the classifier
- k-nearest-neighbours is a *lazy* classifier
- Lazy means the **prediction** is based on the training data only - a separate model is not trained or used for prediction
- The knn algorithm is conceptually simple - vote for the most common label in the local neighbourhood of the unseen data instance
- There are some options in how the classifier is configured, notably choice of distance function and the hyperparameter k
- Classification performance is assessed using a *confusion matrix*
- Predictions do not always match unseen data - ideally they would
- Performance on unseen data (used for testing) is the most common quality criterion, but there are others
- Later, we will meet some metrics that can be computed from the confusion matrix and used to compare different classifiers
- We can use such metrics to validate our classifier choice and search for optimal hyperparameters (*hyperparameter tuning*)